# Report on Student Placement Data Using Percentiles

This report analyses students' academic percentages and salary packages using the 25th percentile (Q1), 50th percentile (Q2), 75th percentile (Q3), 99th percentile, and maximum values (Q4).

| | sl_no | ssc_p | hsc_p | degree_p | etest_p | mba_p | salary |
|---|---|---|---|---|---|---|---|
| **Mean** | 108.0 | 67.303395 | 66.333163 | 66.370186 | 72.100558 | 62.278186 | 288655.405405 |
| **Median** | 108.0 | 67.0 | 65.0 | 66.0 | 71.0 | 62.0 | 265000.0 |
| **Mode** | 1 | 62.0 | 63.0 | 65.0 | 60.0 | 56.7 | 300000.0 |
| **Q1:25%** | 54.5 | 60.6 | 60.9 | 61.0 | 60.0 | 57.945 | 240000.0 |
| **Q2:50%** | 108.0 | 67.0 | 65.0 | 66.0 | 71.0 | 62.0 | 265000.0 |
| **Q3:75%** | 161.5 | 75.7 | 73.0 | 72.0 | 83.5 | 66.255 | 300000.0 |
| **99%** | 212.86 | 87.0 | 91.86 | 83.86 | 97.0 | 76.1142 | NaN |
| **Q4:100%** | 215.0 | 89.4 | 97.7 | 91.0 | 98.0 | 77.89 | 940000.0 |

## 1. Academic performance analysis

### SSLC percentage (`ssc_p`)

- 25th percentile (Q1): 25% of students scored approximately 60.6% or below in SSLC.
- 50th percentile (Q2): 50% of students scored 67% or below.
- 75th percentile (Q3): 75% of students scored 75.7% or below.
- Comparison: Q3 is 15.1 percentage points higher than Q1 (75.7 − 60.6). This shows the spread between the lower and upper quartiles.

### HSC percentage (`hsc_p`)

- 25th percentile: 25% of students scored approximately 60.9% or below.
- 50th percentile: 50% of students scored 65% or below.
- 75th percentile: 75% of students scored 73% or below.
- Comparison: The difference between Q1 and Q3 is 12.1 percentage points (73 − 60.9).

### Degree percentage (`degree_p`)

- 25th percentile: 25% of students scored 61% or below.
- 50th percentile: 50% of students scored 66% or below.
- 75th percentile: 75% of students scored 72% or below.
- Comparison: The difference between Q1 and Q3 is 11 percentage points.

### Entrance test percentage (`etest_p`)

- 25th percentile: 25% of students scored 60% or below.
- 50th percentile: 50% of students scored 71% or below.
- 75th percentile: 75% of students scored 83.5% or below.
- Comparison: The difference between Q1 and Q3 is 23.5 percentage points, the largest spread among the academic measures listed so far.

### MBA percentage (`mba_p`)

- 25th percentile: 25% of students scored approximately 57.95% or below.
- 50th percentile: 50% of students scored 62% or below.
- 75th percentile: 75% of students scored approximately 66.26% or below.
- Comparison: The difference between Q1 and Q3 is approximately 8.31 percentage points.

## 2. Salary package analysis

- 25th percentile (Q1): 25% of students have a salary of approximately ₹2,40,000 or below.
- 50th percentile (Q2): 50% of students have a salary of ₹2,65,000 or below.
- 75th percentile (Q3): 75% of students have a salary of ₹3,00,000 or below.
- Comparison: The difference between Q1 and Q3 is ₹60,000 (₹3,00,000 − ₹2,40,000). The third quartile salary is 25% higher than the first quartile salary.
- Maximum salary: The highest salary shown in the dataset is ₹9,40,000. This is considerably higher than the median salary, suggesting that some students received higher salary packages.